

## CE 310 — Week 1 Weekly Quiz

**Week:** 1 | **Topic:** The Flaw of Averages

**Due:** Closes **8:00 AM Tuesday, Sep 1** — the day before Exercise 1 and Homework 1 are due. Taken on D2L

**Points:** 15 questions × 1 pt each = 15 pts

**Attempts:** Two — the higher score counts

**Time limit:** 25 minutes

**Instructions:** Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

**This quiz covers the Monday lecture and the Wednesday Excel tutorial** — the flaw of averages, percentiles and the interquartile range, sample versus population standard deviation, and the Excel functions and range habits that go with them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

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### Questions

**Q1.** When the U.S. Air Force measured 10 physical dimensions of thousands of pilots to design a cockpit for the “average” pilot, how many pilots were average across all 10 dimensions simultaneously?

- A) Fewer than 100
  - B) About 5% of all pilots
  - C) About 1% of all pilots
  - D) Zero
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**Q2.** You need the 25th percentile, the 75th percentile, and the interquartile range of the 8,760 hourly temperatures. Which statement is correct?

- A) =PERCENTILE(B2:B8761, 0.25) returns Q1, and IQR = Q3 – Q1, the middle half
  - B) =PERCENTILE(B2:B8761, 25) returns Q1, and IQR = max – min, the full spread
  - C) =LARGE(B2:B8761, 0.25) returns Q1, and IQR = Q3 – Q1 across the middle half
  - D) =PERCENTILE(B2:B8761, 0.25) returns Q1, and IQR is the 75th percentile itself
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**Q3.** Column A holds a date-time for each hour and column B the temperature. A helper column E computes =MONTH(A2). You want the average temperature across July hours only. Which formula returns it?

- A) =AVERAGE(B2:B8761)
  - B) =AVERAGEIF(B2:B8761, 7, E2:E8761)
  - C) =AVERAGEIF(E2:E8761, 7, B2:B8761)
  - D) =COUNTIF(E2:E8761, 7)
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**Q4.** [Ordering] ASHRAE Standard 169 defines the cooling design dry-bulb temperature as the value exceeded by the hottest 1% of annual hours. Put the steps in the order you would carry them out.

- (A) Multiply 8,760 hours by 1% to get the rank you need — 88
  - (B) Confirm the dataset covers a full year, so the hour count is 8,760
  - (C) Return the 88th-largest temperature in the column with =LARGE(range, 88)
  - (D) Use that temperature as the design condition for sizing the cooling system
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**Q5.** Tucson's 8,760 hourly temperatures have a mean near 72 °F and a standard deviation near 18 °F, so the coefficient of variation is about 25%. Using the three CV thresholds from lecture, what follows for design?

- A) Below 10% — the mean on its own is a reliable design value
  - B) Between 10% and 20% — quote the mean, but report the standard deviation beside it
  - C) Above 20% — the mean is misleading on its own, so size from a percentile of the distribution instead
  - D) CV only lets you compare spread between different variables; it carries no design implication
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**Q6.** You need the number of hours whose temperature was **at or above 80 °F and strictly below 90 °F**. Which formula returns it?

- A) =COUNTIFS(B2:B8761, ">=80", B2:B8761, "<90")
  - B) =COUNTIF(B2:B8761, ">=80")
  - C) =COUNTIF(B2:B8761, ">=80") + COUNTIF(B2:B8761, "<90")
  - D) =COUNTIFS(B2:B8761, ">=80", "<90")
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**Q7.** A structural engineer designs a roadside billboard using the average annual wind speed to calculate design wind pressure. Compared to the actual peak wind speed, this approach is likely to:

- A) Overestimate wind pressure, making the design unnecessarily expensive
  - B) Accurately represent the worst-case wind load the billboard will experience
  - C) Have no effect, since wind pressure depends on factors other than wind speed
  - D) Underestimate wind pressure, risking structural failure during a storm
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**Q8.** [True/False] Dividing by  $(n - 1)$  instead of  $n$  makes the sample standard deviation slightly larger, correcting a bias that would otherwise cause a sample to understate the true variability of the population it came from.

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**Q9.** [Matching] Match each Excel function to what it returns.

Choices:

- (A) =LARGE ( range, k )
- (B) =COUNTIF ( range, ">100" )
- (C) =SMALL ( range, k )
- (D) =AVERAGE ( range )
- (E) =STDEV ( range )
- (F) =MEDIAN ( range )

Matches:

1. How many values in the range exceed 100
  2. The arithmetic mean of the range
  3. The k-th largest value in the range
  4. The sample standard deviation of the range
  5. The middle value once the range is sorted
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**Q10.** [Select all that apply] An engineer sizes a system using only the annual average outdoor temperature. Which of the following does that single number fail to capture **about the year's temperatures**?

- (A) The hottest 1% of hours, which drive peak cooling load
  - (B) Whether this year ran warmer or cooler than the year before, compared on the same basis
  - (C) Rare extreme cold events at the other end of the distribution
  - (D) The typical or central value of the year's temperatures
  - (E) How much individual hours vary around the mean
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**Q11.** You open Week1\_Tucson\_Weather\_2023.csv, build several formulas, and save with Ctrl+S. You reopen the file the next day and every formula is

gone — only the numbers remain. What happened?

- A) Excel recalculated on open and replaced each formula with the value it produced
  - B) The formulas were dropped because the file exceeds Excel's row limit for CSVs
  - C) The file is still a .csv, and that format stores values only, never formulas
  - D) Excel cleared the formulas because the downloaded source file opened read-only
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**Q12.** You write explicit ranges like B2:B8761 rather than the whole column B:B. Column B has the header text Temp\_F in row 1. Why does it matter?

- A) Whole-column references such as B:B are not permitted inside Excel formulas
  - B) AVERAGE silently ignores the text header, so a careless range still looks right
  - C) A whole-column reference returns #VALUE! whenever the column it spans holds text
  - D) B:B recalculates more slowly than B2:B8761, though the answer is identical
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**Q13.** =COUNTIFS(B2:B8761, ">80", D2:D8760, ">50") returns #VALUE!. What is wrong with it?

- A) The second range is one row short — D2:D8760 against B2:B8761
  - B) The two criteria cannot both be greater-than comparisons in one COUNTIFS
  - C) COUNTIFS accepts only one range/criterion pair; use it twice and add up
  - D) The criteria must be typed as bare numbers here, not as text in quotation marks
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**Q14.** You type =COUNTIF(B2:B8761, >95) to count the hours above 95 °F and Excel returns #VALUE!. Your neighbour's =COUNTIF(E2:E8761, 7) counts July hours with no quotation marks at all. What is the rule?

- A) Every COUNTIF criterion must be quoted; the unquoted 7 works only because column E holds whole numbers
- B) Comparison operators are not allowed in COUNTIF — use COUNTIFS whenever you need > or <
- C) A criterion that uses a comparison operator must be quoted — ">95" — while a plain value like 7 needs no quotes

- D) Quotes are never required in COUNTIF; the #VALUE! comes from the range including the header row
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**Q15.** You put =MONTH(A2) in cell E2 and fill it down. You then press Ctrl+End to check your work. What should cell E8761 contain?

- A) 1 — the fill copied the formula exactly, so every row still reports January
- B) 12 — the reference adjusts row by row, so the last row reports December
- C) 8760 — the fill numbered the rows sequentially instead of reading the date
- D) Anything at all; AVERAGEIF re-reads column A, so column E is never consulted